CARA AUSTRING

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EDUCATION

University of Southern California — Iovine and Young Academy (IYA) *Los Angeles, CA*

B.S., Business of Innovation | Freshman, Class of 2029

DESIGN & TECHNICAL PROJECTS

AR Gesture-Prediction Interface (Brain-Computer Interface) *August 2025*

- Designed and animated a hand-gesture interaction in Spline, translating raw neural signals into an intuitive AR visualization for the user
- Built a Python classification pipeline connecting a Neurosity Crown EEG headset to the AR experience, enabling it to accurately predict which hand a user was about to raise after a short calibration period

W17 Vehicle Design, Prototype & Motorized Track *2024 – 2025*

- Designed a W17 vehicle from concept to physical prototype, modeling the full form in OnShape and 3D printing the final build
- Iterated on the mechanical design to integrate a drive motor and custom track, turning a static model into a working interactive product

ESP32 Scavenger Hunt Experience *2024*

- Designed the end-to-end participant experience for a location-based scavenger hunt across downtown Summit, NJ, including a custom 3D-printed suitcase enclosure for the hardware
- Programmed proximity-based interaction logic on ESP32 microcontrollers so each step felt like a natural, rewarding progression — driving foot traffic for local partner businesses

PROFESSIONAL EXPERIENCE

Innovation Foundry — Founder *August 2022 – Present*

Summit, NJ (currently managed remotely while attending USC)

- Designed and continuously iterated a 3D printing curriculum for ~275 unique customers over 4 years, many repeat students, refining lessons based on feedback and engagement across classes (capped at 12) and sold-out 9-week summer camps (capped at 15) for ages 8–12
- Recruited, trained, and managed a team of 1 full-time and up to 12 seasonal employees; continue to guide curriculum and business strategy remotely from USC while mentoring a family successor as on-site lead

Women in STEM Program — Founder & Lead Instructor *September 2025 – Present*

- Designed and launched an AR/VR-focused curriculum for young girls, translating complex technical concepts into approachable, hands-on lessons
- Owned end-to-end program design — budget, scheduling, curriculum, scholarships, and staffing — as lead instructor alongside a second employee

LEADERSHIP & CREATIVE EXPERIENCE

Stage Crew — Set Designer *September 2022 – Present*

- Designed and built multi-part sets for school theatrical productions, balancing visual storytelling with technical and structural constraints

TSA Engineering Club — Participant *September 2024 – Present*

- Co-designed and prototyped a hurricane- and flood-resilient hotel structure for a national competition; communicated design decisions in a co-authored technical essay and advanced to TSA Nationals

Crochet Club — President *September 2022 – Present*

- Designed original patterns and a beginner-friendly curriculum, growing club membership through creative programming and fundraising events

SKILLS

Design: Spline (AR/3D Prototyping), OnShape (CAD), 3D Modeling & Animation, Instructional & Curriculum Design, Visual & Spatial Design, Storytelling

Technical: Python, ESP32 / Embedded Systems, 3D Printing, Team Leadership, Time Management